Production RAG Parameter Guide

Standard starting values for each stage of a RAG pipeline, with the significance of each parameter. These are starting points, not universal answers: always validate against your own evaluation set.

1. Chunking

Parameter	Standard value	Significance
Chunk size	256-512 tokens (factual Q&A); 512-1024 (technical or narrative docs)	Smaller chunks give precise matches but lose context. Larger chunks keep context but dilute the embedding with unrelated text.
Chunk overlap	10-20% (50-100 tokens)	Prevents an idea from being split across a boundary. Too much inflates the index and returns near-duplicates.
Splitter	Recursive (paragraph, sentence, word) or structure-aware (headings, tables, code blocks)	Respects natural boundaries so chunks stay coherent. Fixed-size character splitting is the weakest option.
Parent-child chunks	Child 128-256, parent 1024-2048	Search on small chunks for precision, then pass the larger parent to the LLM for context.
Metadata per chunk	source, title, section, date, doc_id, ACL tags	Enables filtering, citations, access control, and freshness handling.

2. Embedding

Parameter	Standard value	Significance
Model	text-embedding-3-small/large, Cohere embed v3, BGE, E5, GTE	The biggest single driver of retrieval quality. Use the same model for indexing and querying.

Parameter	Standard value	Significance
Dimensions	768-1536 (truncate if the model supports it)	Higher dimensions capture more nuance but cost more memory and latency.
Normalize vectors	true	Required for cosine/dot-product equivalence and consistent scores.
Batch size	64-256	Indexing throughput, limited by API rate limits or GPU memory.
Query/document prefixes	Model-specific (E5 and BGE need them)	Skipping required prefixes silently degrades quality.

3. Vector Index (ANN)

Parameter	Standard value	Significance
Distance metric	cosine	Must match how the embedding model was trained.
HNSW M	16 (up to 32-64 for high recall)	Graph links per node. Higher gives better recall but more memory.
HNSW ef_construction	200 (128-512)	Build-time search depth. Higher gives a better index but slower builds.
HNSW ef_search	64-128 (must be at least top-k)	Query-time depth. The main recall vs. latency dial.
IVF nlist / nprobe	nlist about sqrt(N); nprobe 8-32	Number of clusters, and how many are searched per query.

4. Retrieval

Parameter	Standard value	Significance
Candidate top-k	20-50 (with reranking); 4-8 (without)	Wide net at stage one. Too low misses the answer; too high adds latency and cost.

Parameter	Standard value	Significance
Search mode	Hybrid (dense + BM25)	Dense handles meaning; BM25 handles exact terms like IDs, names, and error codes.
Hybrid weight (alpha)	0.5-0.7 toward dense	Balance between semantic and keyword matching. Tune by query type.
Fusion method	Reciprocal Rank Fusion, k=60	Merges rankings without needing comparable scores.
Similarity threshold	About 0.3-0.5 for cosine (calibrate per model)	Drops weak matches so the LLM isn't fed noise, and enables "no answer found" behavior.
MMR lambda / fetch_k	0.5-0.7 / 20-40	Balances relevance against diversity, avoiding near-identical chunks.
Metadata filters	Applied pre-search	Narrows by tenant, date, or doc type. Essential for multi-tenant security.

5. Reranking

Parameter	Standard value	Significance
Reranker	Cross-encoder (bge-reranker, Cohere Rerank)	Scores query and chunk jointly, far more accurate than vector similarity alone. Often the largest quality gain after hybrid search.
Input candidates	30-50	Reranking cost scales linearly with this number.
Final top-n	3-8	What actually enters the prompt. More context is not better; models lose focus in long contexts.
Min rerank score	Calibrated on your data	A second quality gate before generation.

6. Query Processing

Parameter	Standard value	Significance
Query rewriting	On for conversational apps	Resolves pronouns and follow-ups into standalone queries.
Multi-query expansion	3-4 variants	Improves recall on ambiguous questions at the cost of extra LLM calls.
HyDE	Optional	Helps when queries are short and documents are long. Adds latency.

7. Generation (LLM)

Parameter	Standard value	Significance
Temperature	0-0.2	Low randomness keeps answers faithful to the context.
Top-p	1.0 (or 0.9)	Adjust temperature or top-p, not both.
Max output tokens	500-1000	Caps cost and runaway answers.
Context budget	Retrieved text at most 50-60% of the window	Leaves room for the system prompt, history, and the answer.
System prompt	"Answer only from the provided context; if it is not there, say you don't know; cite sources."	The strongest hallucination control you have.
Timeout / retries	30-60s / 2-3 with backoff	Resilience against provider hiccups.

8. Example Configuration

```
chunking:
  size: 512
  overlap: 80
  splitter: recursive
embedding:
  model: text-embedding-3-large
```

```
  dimensions: 1024
  normalize: true
index:
  metric: cosine
  hnsw: {M: 32, ef_construction: 200, ef_search: 100}
retrieval:
  mode: hybrid
  alpha: 0.6
  fusion: rrf
  rrf_k: 60
  candidate_k: 40
  score_threshold: 0.35
  mmr: {lambda: 0.6, fetch_k: 40}
reranker:
  model: bge-reranker-v2-m3
  top_n: 5
  min_score: 0.2
generation:
  temperature: 0.1
  top_p: 1.0
  max_tokens: 800
  context_budget_ratio: 0.5
```

9. Production Practices

- **Evaluate continuously.** Track retrieval recall@k (aim for above 90% on your test set), MRR, and answer faithfulness/relevance (for example with RAGAS).
- **Change one parameter at a time** and re-run the same test set.
- **Cache** embeddings and frequent queries.
- **Monitor** p95 latency, cost per query, and the "no answer" rate.
- **Version your index.** Changing the embedding model or chunking requires a full re-index.
- **Enforce access control** at retrieval time through metadata filters, not in the prompt.

10. Suggested Tuning Order

1. Chunk size and overlap
2. Hybrid search and alpha
3. Candidate top-k and reranker top-n
4. Similarity and rerank thresholds
5. System prompt and generation settings

6. ANN index parameters (ef_search, M) for latency/recall trade-offs